

Lightweight IoT Security Processor

Questa waveform report for thesis submission

SIMULATION VCD

simulation\questa\riscv_core_tb.vcd

SIMULATION WINDOW

0 ns to 15205.0 ns

REGRESSION RESULT

PASS 0x00000048, FAIL 0x00000000

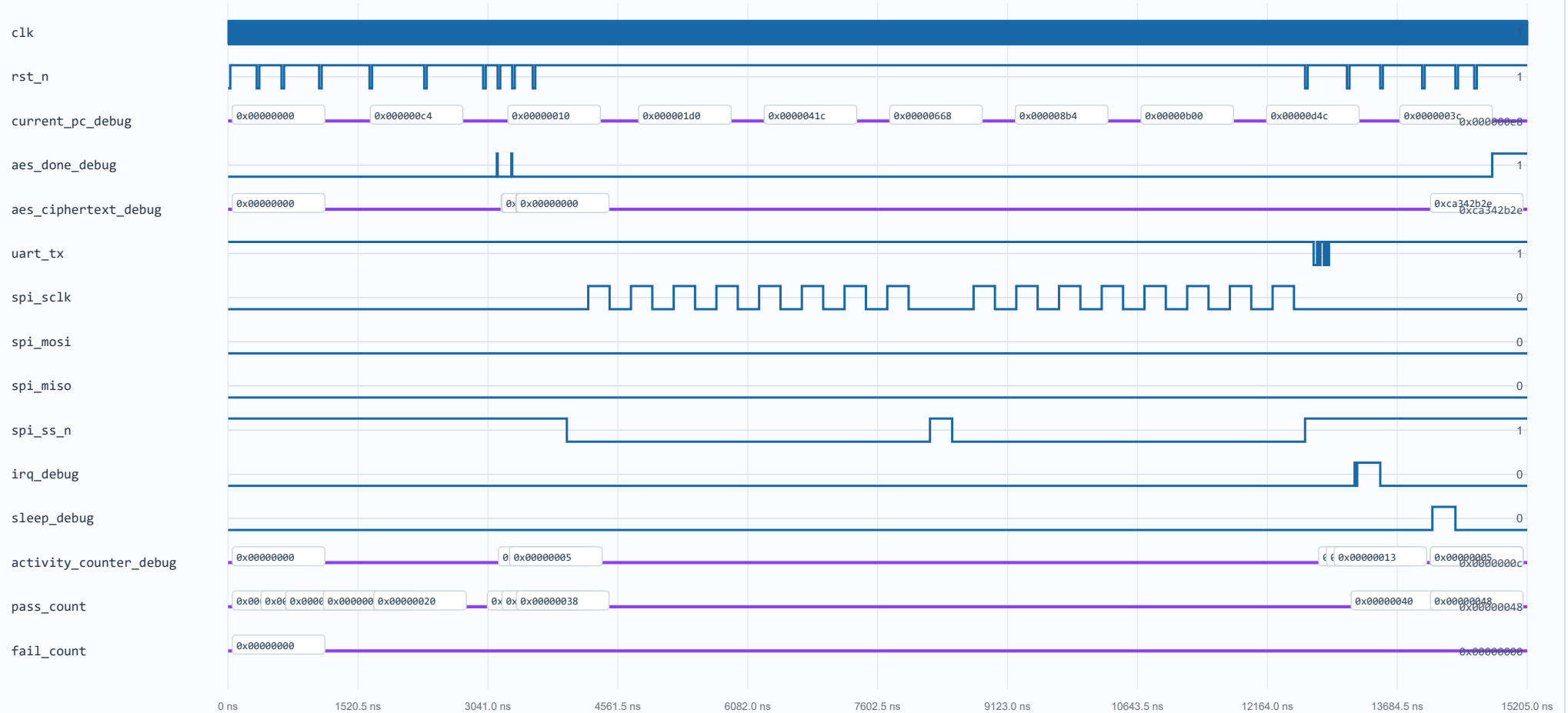
GENERATED

21/5/2026, 9:51:19 pm

The pages group the important waveforms from the full VCD: top debug outputs, CPU pipeline, MMIO decode, AES ECB/CTR, Sensor/SPI IP, UART, interrupt controller, DMA-lite, and power/activity counters.

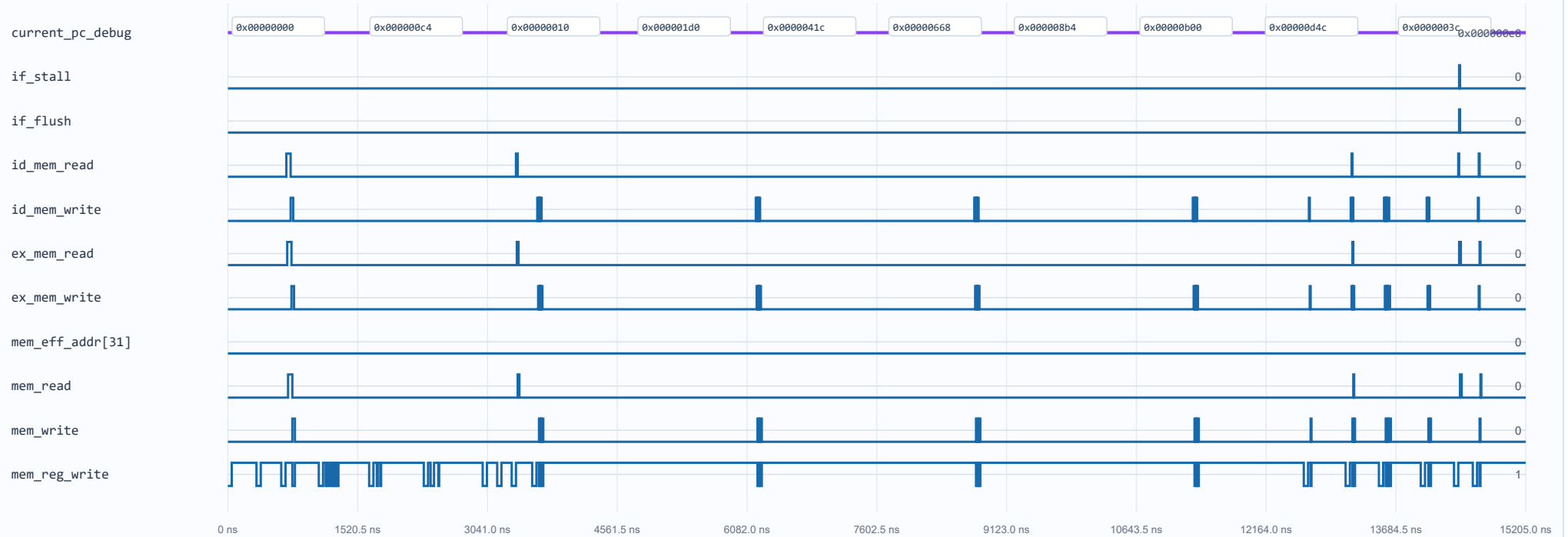
1. Verification Overview and Top-Level Debug Outputs

Full simulation result signals and FPGA-visible debug outputs from the Lightweight IoT Security Processor top level.



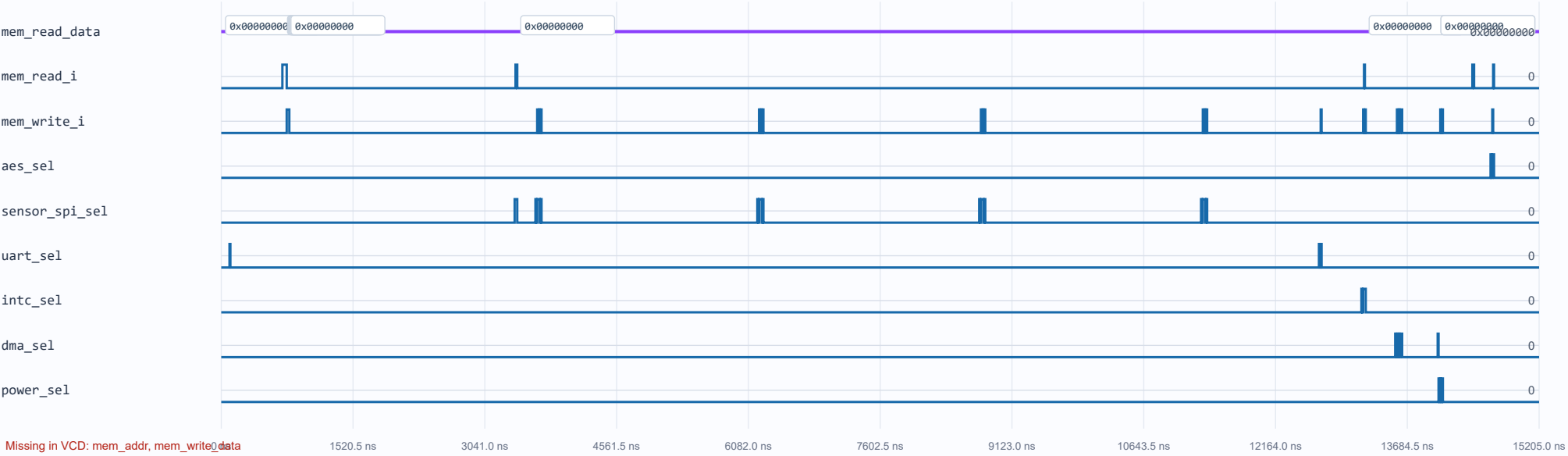
2. CPU Pipeline and Hazard Control

Instruction flow, PC movement, stall and flush activity, and memory-stage control.



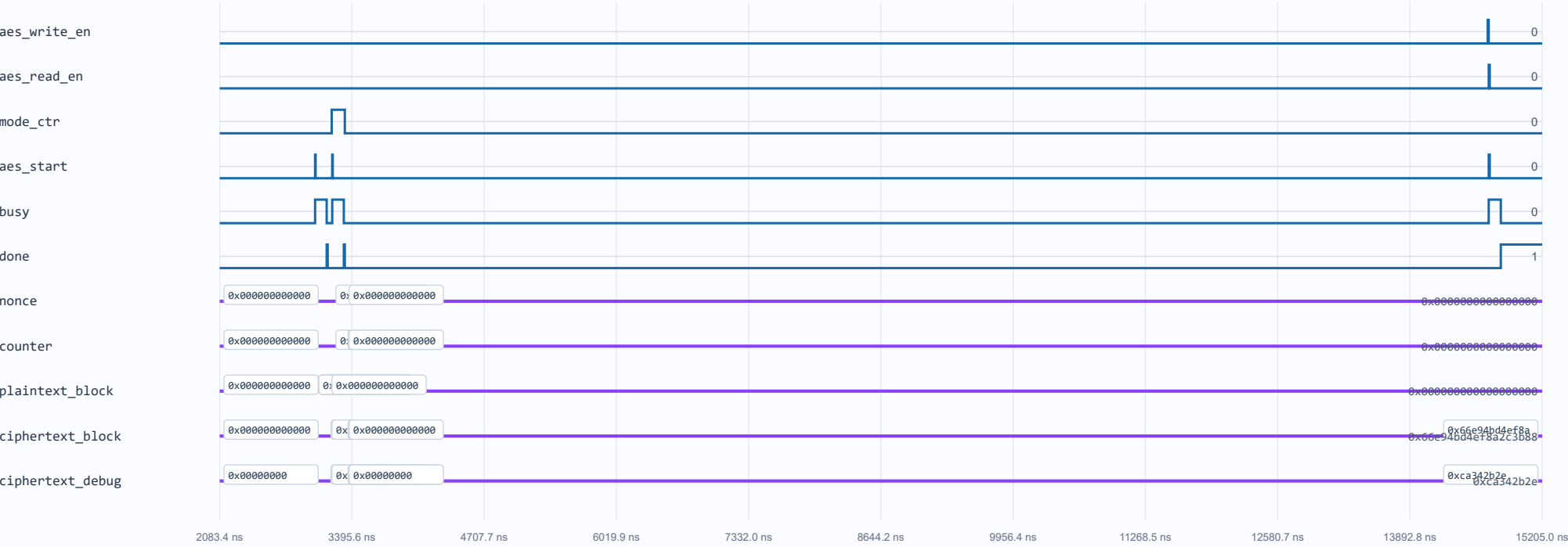
3. MMIO Decode and Peripheral Routing

Address decode selects exactly one MMIO target while preserving normal data memory behavior.



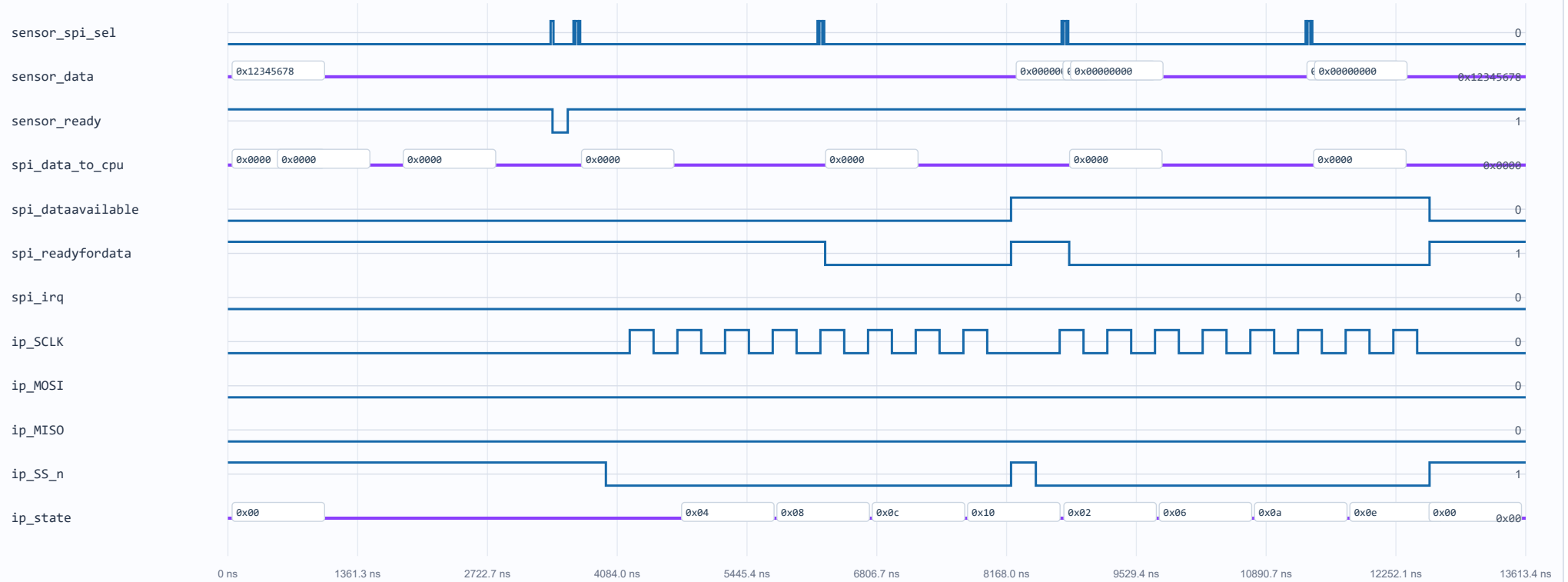
4. AES-128 ECB and CTR Accelerator

AES primitive remains intact; CTR mode uses nonce/counter as the encrypted keystream block and XORs plaintext into ciphertext.



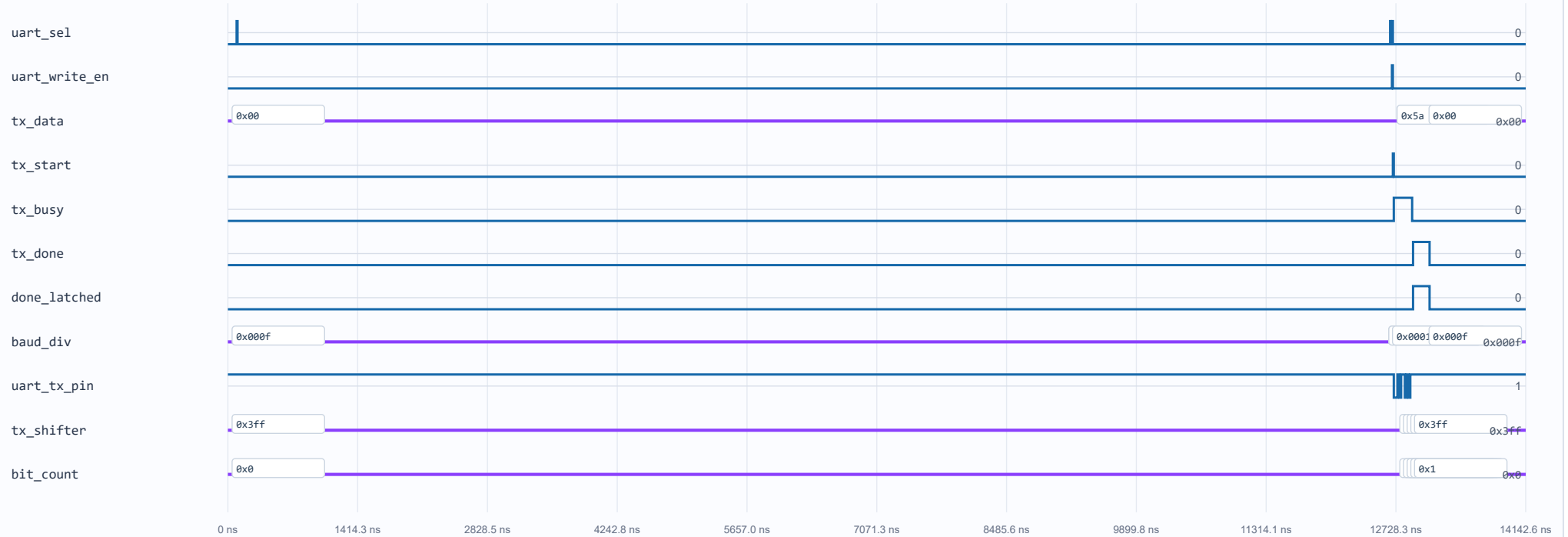
5. Sensor MMIO and Intel Avalon SPI IP

Sensor registers are retained while the SPI IP window provides real serial I/O signals.



6. UART Encrypted Output Peripheral

CPU store to UART TX data launches serial transmission; busy and done flags are visible to software and interrupts.



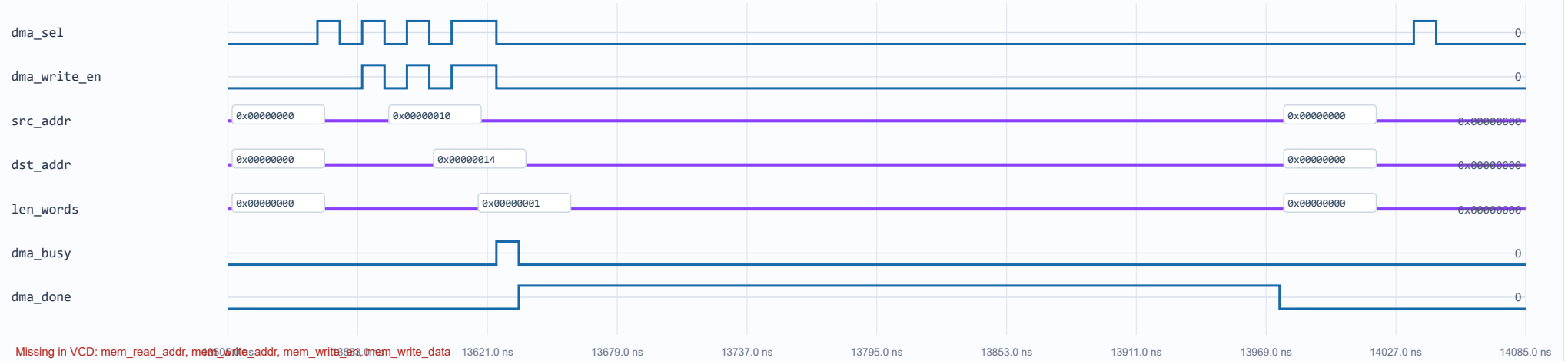
7. Interrupt Controller

AES done, UART done, sensor ready and DMA done events are latched into pending bits and gated by enable bits.



8. DMA-Lite Transfer Engine

Word-transfer DMA-lite control/status and its internal data-mem side channel.



9. Low-Power Sleep and Activity Counters

Clock enables are used for activity reduction, and counters support the measurement of CPU/peripheral activity.

